Astronomy - Graduate Level Assessment 25 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the significance of the 1:2:4 resonance in the Jupiter's moons system?
   (a) The resonance pulls Io in different directions and generates heat.
   (b) It makes the orbit of Io slightly elliptical.
   (c) It creates a gap with no asteriods between the orbits.
   (d) It prevents formation of the ring material into other moons.
2. In addition to the conditions required for any solar eclipse what must also be true in order for you to observe a total solar eclipse?
   (a) The Earth must lie completely within the Moon's penumbra.
   (b) The Moon's penumbra must touch the area where you are located.
   (c) The Earth must be near aphelion in its orbit of the Sun.
   (d) The Moon's umbra must touch the area where you are located.
3. Approximately how long does it take Pluto to orbit the Sun once?
   (a) 150 years
   (b) 200 years
   (c) 250 years
   (d) 300 years
4. Which factor s most important in determining the history of volanism and tectonism on a planet?
   (a) size of the planet
   (b) presence of an atmosphere
   (c) distance from the sun
   (d) rotation period
5. Which of the following is/are NOT caused by orbital resonance?
   (a) 2:3 periodic ratio of Neptune:Pluto
   (b) Kirkwood Gaps.
   (c) Gaps in Saturn's rings.
   (d) Breaking of small Jovian moons to form ring materials.

True / False (10 marks)

Answer True or False

6. True or False: The Cassini division is a prominent gap in Saturn's rings, formed by the gravitational influence of the moon Mimas.

7. True or False: A formation theory of the solar system does not need to explain the presence of life on Earth.

8. True or False: The Moon's rotational period is equal to its orbital period, resulting in the same face always facing the Earth.

9. True or False: Saturn and Jupiter do not share an equatorial wind speed exceeding 900 miles per hour.

10. True or False: The Arecibo Telescope, which was operational until 2020, had the largest parabolic antenna among radio telescopes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of referring to Jupiter and other jovian planets as "gas giants," particularly in relation to their non-gaseous components and overall composition.

12. Discuss the role of hydrogen in the composition and atmospheric dynamics of the Jovian planets, highlighting its significance compared to other elements.

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